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Java SE 27 & JDK 27

Class GZIPInputStream

java.lang.Object
  java.io.InputStream
    java.io.FilterInputStream
      java.util.zip.InflaterInputStream
        java.util.zip.GZIPInputStream

All Implemented Interfaces:

`Closeable`, `AutoCloseable`

```
public class GZIPInputStream
extends InflaterInputStream
```

This class implements a stream filter for decompressing GZIP file format data.

GZIP file format

The GZIP file format is specified by RFC 1952. The format, as specified in section 2.2 of the RFC, consists of a series of "members" that appear one after another in the stream with no additional information before, between, or after them. Each member consists of a header, followed by data that is compressed using the `deflate` algorithm, and then a trailer.

This class is capable of reading a stream consisting of a series of members.

Reading from the stream may read and buffer bytes from the underlying stream. This includes bytes that follow a member's trailer. Whether or not any additional bytes have been read past a member's trailer, the read methods on this class yield decompressed data from at most one member; data from multiple members is not combined in a single read operation.

Thread safety

`GZIPInputStream` is not safe for use by multiple concurrent threads. Any multithreaded concurrent use must be guarded by appropriate synchronization.

API Note:

The `close()` method should be called to release resources used by this stream, either directly, or with the `try`-with-resources statement.

Implementation Note:

After reading a member trailer, the read method calls `InputStream.available()` on the underlying stream to determine whether additional bytes are available that may represent a subsequent member. If the `jdk.util.gzip.tryReadAheadAfterTrailer` system property is set to `true`, then the call to `InputStream.available()` is skipped and the

implementation instead attempts to read a subsequent member in the stream. `GZIPInputStream` depends on the return value of `InputStream.available()` to reliably process a stream with a series of members. Consequently, it may be necessary to set this property in environments that process streams with a series of members. By default, the `jdk.util.gzip.tryReadAheadAfterTrailer` system property is not set, and `InputStream.available()` gets called.

Since:

1.1

External Specifications

RFC 1952: GZIP file format specification version 4.3

See Also:

`InflaterInputStream`

Field Summary

Fields

Modifier and Type	Field	Description
`protected CRC32`	**`crc`**	CRC-32 for decompressed data.
`protected boolean`	**`eos`**	Indicates end of input stream.
`static final int`	**`GZIP_MAGIC`**	GZIP header magic number.

Fields declared in class InflaterInputStream

`buf`, `inf`, `len`

Fields declared in class FilterInputStream

`in`

Constructor Summary

Constructors

Constructor	Description
`GZIPInputStream``(InputStream in)`	

	Creates a new input stream with a default buffer size.
`GZIPInputStream(InputStream in, int size)`	Creates a new input stream with the specified buffer size.

Method Summary

All Methods | Instance Methods

Modifier and Type	Method	Description
`void`	`close()`	Closes this input stream and releases any system resources associated with the stream.
`int`	`read(byte[] buf, int off, int len)`	Reads decompressed data into an array of bytes, returning the number of decompressed bytes.

Methods declared in class InflaterInputStream

`available`, `fill`, `mark`, `markSupported`, `read`, `reset`, `skip`

Methods declared in class FilterInputStream

`read`

Methods declared in class InputStream

`nullInputStream`, `readAllBytes`, `readNBytes`, `readNBytes`, `skipNBytes`, `transferTo`

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Details

GZIP_MAGIC

```
public static final int GZIP_MAGIC
```

GZIP header magic number.

See Also:

Constant Field Values

crc

```
protected CRC32 crc
```

CRC-32 for decompressed data.

eos

```
protected boolean eos
```

Indicates end of input stream.

Constructor Details

GZIPInputStream

```
public GZIPInputStream(InputStream in,
                       int size)
                throws IOException
```

Creates a new input stream with the specified buffer size.

Parameters:

`in` - the input stream

`size` - the input buffer size

Throws:

`ZipException` - if a GZIP format error has occurred or the compression method used is unsupported

`NullPointerException` - if `in` is null

IOException - if an I/O error occurs when reading the member header from the underlying stream

IllegalArgumentException - if `size <= 0`

GZIPInputStream

```
public GZIPInputStream(InputStream in)
                throws IOException
```

Creates a new input stream with a default buffer size.

Parameters:

`in` - the input stream

Throws:

ZipException - if a GZIP format error has occurred or the compression method used is unsupported

NullPointerException - if `in` is null

IOException - if an I/O error occurs when reading the member header from the underlying stream

Method Details

read

```
public int read(byte[] buf,
                int off,
                int len)
         throws IOException
```

Reads decompressed data into an array of bytes, returning the number of decompressed bytes. If `len` is not zero, the method will block until some input can be decompressed; otherwise, no bytes are read and `0` is returned.

If this method returns a nonzero integer *n* then `buf[off]` through `buf[off+`*n*`-1]` contain the decompressed data. The content of elements `buf[off+`*n*`]` through `buf[off+`*len*`-1]` is undefined, contrary to the specification of the InputStream superclass, so an implementation is free to modify these elements during the inflate operation. If this method returns `-1` or throws an exception then the content of `buf[off]` through `buf[off+`*len* `-1]` is undefined.

Overrides:

read in class InflaterInputStream

Parameters:

`buf` - the buffer into which the data is read

`off` - the start offset in the destination array `buf`

`len` - the maximum number of bytes to read into `buf`

Returns:

the actual number of bytes decompressed from a GZIP member, or -1 if the end-of-stream is reached

Throws:

`NullPointerException` - If `buf` is `null`.

`IndexOutOfBoundsException` - If `off` is negative, `len` is negative, or `len` is greater than `buf.length - off`

`ZipException` - if the compressed input data is corrupt.

`IOException` - if the stream is closed or an I/O error has occurred.

See Also:

GZIP file format

close

```
public void close()
           throws IOException
```

Closes this input stream and releases any system resources associated with the stream.

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in interface `Closeable`

Overrides:

`close` in class `InflaterInputStream`

Throws:

`IOException` - if an I/O error has occurred

See Also:

`FilterInputStream.in`

Report a bug or suggest an enhancement
For further API reference and developer documentation see the Java SE Documentation, which contains more detailed, developer-targeted descriptions with conceptual overviews, definitions of terms, workarounds, and working code examples. Other versions.
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